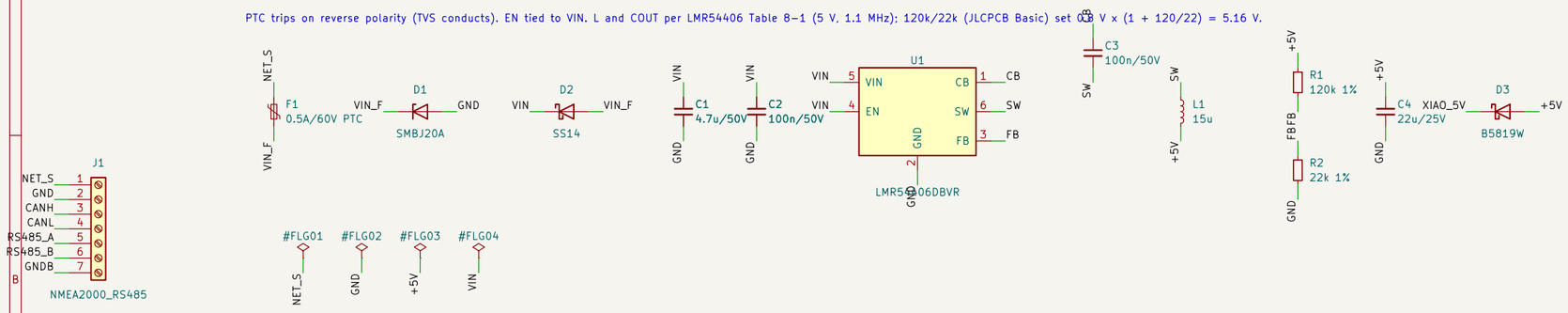
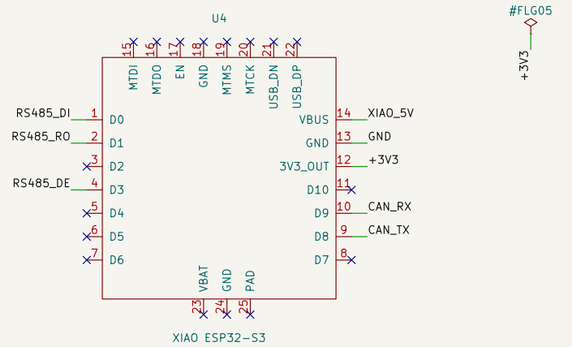
Power: NMEA 2000 drop → 5 V

PTC trips on reverse polarity (TVS conducts). EN tied to VIN. L and COUT per LMR54406 Table 8-1 (5 V, 1.1 MHz): 120k/22k (JLCPCB Basic) set 0.8 V x (1 + 120/22) = 5.16 V.

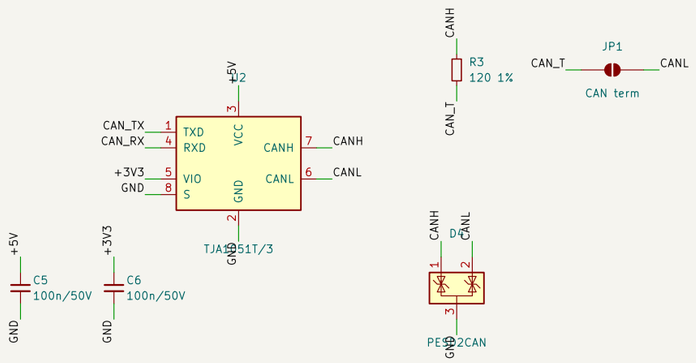


Controller: Seeed XIAO ESP32-S3 (bottom pads not used; keep copper out from under it)



CAN (NMEA 2000): non-isolated, bus-powered

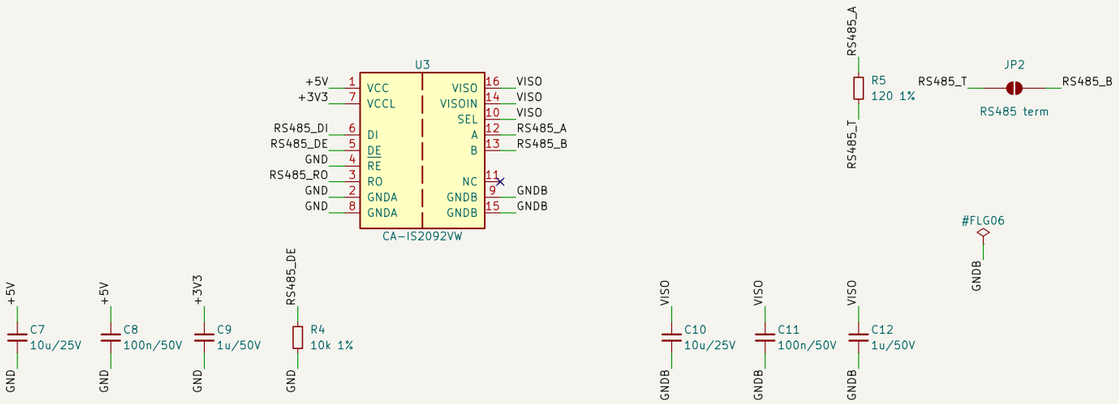
Bus pins ±58 V. S tied low: normal mode. JP1 closed = 120 Ω termination (leave open on a drop).



RS485 / RS422: isolated (CA-IS2092VW, 3.75 kVrms)

VCC from 5 V (powers the isolated DC-DC), VCCL 3.3 V logic. SEL to VISO: VISO = 5 V. RE low: receiver always on.

JP2 closed = 120 Ω termination. GNDB goes to J1 pin 7 for the RS422 device's signal ground.



CERN-OHL-S-2.0
XIAO ESP32-S3 carrier: NMEA 2000 (CAN) + isolated RS485, bus powered

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